Fusion-JEPA:
Expressive, Low-Resource Text-to-Speech via Joint-Embedding Predictive Architecture

Omar Alkhammash, **Abdulrahman Soliman**, **Hassan Alahmed**, **Kerven Durdymyradov**

g3m438@gmail.com, amohamed0057@gmail.com, hassan.yaseen.alahmed@gmail.com, kerventkm@gmail.com

Abstract

We present Fusion-JEPA, an end-to-end neural speech synthesis architecture adapting Denoising with a Joint-Embedding Predictive Architecture (D-JEPA) for continuous acoustic modeling. While designed to overcome the acute data scarcity and morphological complexity of Modern Standard Arabic (MSA), the architecture natively extends to English, providing a robust bilingual framework. Traditional acoustic models trained directly via pixel-level L_p-norm spectrogram regression frequently suffer from spectral oversmoothing and attention collapse in low-resource regimes (< 2 hours). To resolve this, our system decouples abstract phonetic representation learning from acoustic distribution modeling. Operating on 2D log-Mel spectrogram patches (128 frequency bins) with high-ratio random masking and Exponential Moving Average target updates, the model learns invariant speech representations without pixel-level reconstruction. We model speech conditioning via a Multimodal Diffusion Transformer (MM-DiT) equipped with 1D Rotary Position Embeddings (1D RoPE) and Conditional Flow Matching ($\mathcal{L}_v$), synthesized into 44.1kHz audio via BigVGAN neural vocoding. On isolated overfitting verification benchmarks across both English (LJSpeech) and Arabic (Nawar Halabi's Arabic Speech Corpus), our architecture successfully achieves intelligible speech generation and accurate harmonic formant reconstruction, validating the stability and generative capacity of joint-embedding flow matching for low-resource speech synthesis. The code is now available: `https://github.com/OmarA32/Fusion-JEPA-TTS`.

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Code: `https://github.com/OmarA32/Fusion-JEPA-TTS`

Project Page: `https://omara32.github.io/Fusion-JEPA-TTS/`

Contents

1 Introduction

Text-to-Speech (TTS) synthesis has achieved exceptional fidelity for high-resource languages such as English. However, developing high-quality, natural TTS systems for Modern Standard Arabic (MSA) presents distinct computational and linguistic challenges. Written Arabic is largely unvocalized (lacking short vowel diacritics), creating pronunciation ambiguities where identical grapheme sequences map to different phonetic realizations depending on context. Furthermore, Arabic speech synthesis is severely constrained by data scarcity; high-fidelity single-speaker studio datasets are limited in volume, while crowdsourced multilingual corpora contain acoustic noise and microphone variations that destabilize prosodic modeling.

When conventional acoustic models—such as autoregressive sequence decoders or direct L_1/L_2 spectrogram regressors—are trained on scarce paired audio ($<$ 4 hours), they frequently suffer from representation collapse, attention misalignment, and spectral oversmoothing. Optimizing directly on pixel intensities causes models to predict the conditional mean of possible phonetic realizations, blurring high-frequency consonants and guttural transitions.

To address these limitations, we propose decoupling *self-supervised representation learning* from *continuous acoustic distribution modeling*. Inspired by Denoising with a Joint-Embedding Predictive Architecture (`D-JEPA`) [1], we investigate the following core research question: *Can joint-embedding predictive representations coupled with continuous flow matching enable stable, high-fidelity Arabic and English speech synthesis under constrained data conditions?*

1.1 Problem Formulation and Assumptions

The objective of this work is to design and validate an end-to-end non-autoregressive speech synthesis architecture capable of synthesizing coherent 44.1kHz audio from phonetic text. Our approach is guided by two foundational design principles:

1. **Latent Semantic Prediction over Pixel Reconstruction:** Rather than forcing the encoder to reconstruct exact time-frequency intensities, predicting abstract latent representations in an EMA-updated momentum space forces the network to capture invariant phonetic structures without overfitting to high-frequency acoustic noise.
2. **Continuous Flow-Matching Diffusion:** Rather than using complex external pitch or duration extractors, modeling per-token acoustic velocity trajectories via Conditional Flow Matching ($\mathcal{L}_v$) enables end-to-end continuous generation with sharp formant boundaries.

1.2 Summary of Proposed System and Contributions

We introduce **Fusion-JEPA**, an end-to-end neural speech synthesis architecture that scales joint-embedding predictive learning to continuous audio generation. Our development progressed through two major engineering milestones:

- **Initial Adaptation (JEPA-T with SpatialDiT):** We translated the image-based `JEPA-T` framework [2] to 2D log-Mel spectrograms, incorporating a dedicated spatial transformer diffuser (`SpatialDiT`) conditioned on cross-attended phoneme features.
- **D-JEPA Multimodal Overhaul (v11):** To eliminate cross-attention bottlenecks and accommodate variable sentence lengths, we upgraded the core backbone to the **D-JEPA** architecture [1], integrating **Multimodal Diffusion Transformer (MM-DiT)** joint attention blocks [3], **1D Rotary Position Embeddings (1D RoPE)** [4], direct phoneme embedding lookup, and 128 mel frequency bins.

Key Insight

Core Architectural Shift: Unlike traditional speech models that average acoustic variations via pixel-level reconstruction, the D-JEPA framework combines self-supervised latent prediction with continuous flow matching. The diffusion objective prevents latent collapse, while the JEPA objective provides the generative model with high-level phonetic guidance.

The primary contributions of this work are:

- **Adaptation of D-JEPA and MM-DiT for Speech Synthesis:** We implement the first TTS architecture combining D-JEPA joint-embedding prediction with bidirectional MM-DiT multimodal attention and 1D RoPE, enabling mutual conditioning between text phonemes and acoustic representations.
- **End-to-End Flow Matching Without Explicit Alignment Extractors:** We eliminate auxiliary pitch and duration extractors by allowing the flow-matching velocity field to organically learn phonetic-to-acoustic alignments.
- **Empirical Validation with Working Bilingual Audio:** We validate the system on overfitting benchmarks across both English (`LJSpeech`) and Arabic (Nawar Halabi's Arabic Speech Corpus), generating intelligible, high-fidelity speech waveforms with `BigVGAN` at 44.1kHz.

2 Related Work

Our research builds upon three core developments: self-supervised joint-embedding predictive learning, multimodal diffusion transformers with flow matching, and neural vocoding.

2.1 Joint-Embedding Predictive Architectures and D-JEPA

Self-supervised representation learning was fundamentally advanced by the Joint-Embedding Predictive Architecture (JEPA) proposed by LeCun [5]. Unlike pixel-reconstruction methods such as Masked Autoencoders (MAE), `I-JEPA` [6] predicts masked target representations directly in an abstract latent space using an Exponential Moving Average (EMA) teacher network.

Recently, Chen et al. [1] introduced **D-JEPA**, pioneering the integration of JEPA representation learning within generative modeling. D-JEPA demonstrated that applying a per-token diffusion or flow matching loss alongside a Smooth L_1 latent prediction loss naturally prevents representation collapse in the energy landscape without requiring contrastive pairs or complex teacher warmup schedules. Concurrently, `JEPA-T` [2] explored text-conditioned latent prediction for image synthesis. In this work, we build directly upon D-JEPA, translating its continuous generative formulation into speech synthesis for Arabic and English.

2.2 Multimodal Diffusion Transformers and Flow Matching

In generative modeling, continuous Flow Matching [7] has emerged as an efficient, stable alternative to standard discrete diffusion, modeling probability trajectories along straight vector fields between Gaussian noise and data distributions.

To fuse multimodal signals (such as text and continuous latents) without information bottlenecks, Esser et al. [3] introduced the **Multimodal Diffusion Transformer (MM-DiT)** for Stable Diffusion 3. MM-DiT maintains separate LayerNorm and MLP parameters for text and target modalities while concatenating keys and values (K_{joint}, V_{joint}) during attention, enabling dynamic bidirectional conditioning. Furthermore, to accommodate varying sequence lengths without fixed grid distortion, Su et al. [4] formulated **1D Rotary Position Embeddings (RoPE)**. We integrate

MM-DiT blocks and 1D RoPE into the D-JEPA acoustic backbone to handle variable-length speech utterances naturally.

2.3 Speech Corpora and Neural Vocoding

For acoustic modeling, single-speaker studio recordings provide the clean acoustic targets essential for prosodic stability. In English, the `LJSpeech` dataset [8] (13,100 studio clips, $\sim$ 24 hours) serves as the gold-standard benchmark. For Arabic, Nawar Halabi's Arabic Speech Corpus (ASC) [9] provides 1,813 fully annotated Modern Standard Arabic utterances recorded in a professional studio setting.

To convert continuous generated spectrograms into audible time-domain waveforms, neural vocoders are required. **BigVGAN v2** [10] provides universal, high-fidelity vocoding at 44.1kHz using periodic Snake activations and anti-aliased convolutions. We standardize our acoustic outputs on 128 mel frequency bins to maintain native mathematical alignment with BigVGAN's universal pretrained generator.

3 Methodology

The **Fusion-JEPA** architecture integrates joint-embedding representation learning with continuous flow matching for end-to-end speech synthesis. The system operates on 2D log-Mel spectrograms, processing acoustic and linguistic features in a unified multimodal transformer backbone.

3.1 Architectural Formulation

Let $X \in \mathbb{R}^{1\times H\times W}$ denote a log-Mel spectrogram with $H = 128$ frequency bins and $W = 512$ temporal frames. Following standard Vision Transformer (ViT) patchification with square patch size $P = 16$, the spectrogram is segmented into a non-overlapping token grid of dimension $N_h = 128/16 = 8$ and $N_w = 512/16 = 32$, yielding a sequence length $N = N_h \times N_w = 256$ tokens. Each patch is linearly projected into an embedding space of dimension $D = 768$:

$$U = \text{Linear}(\text{Patchify}(X)) \in \mathbb{R}^{N\times D} \tag{1}$$

As illustrated in Figure 1, the architecture comprises three main transformer components:

1. **Context Encoder (ϕ):** Processes the visible (unmasked) acoustic tokens $Y \subset U$ alongside linguistic phoneme embeddings C.
2. **Target Encoder ($\bar{\phi}$):** An Exponential Moving Average (EMA) teacher that encodes all unmasked tokens U, producing target representations $g_i = \text{sg}(\bar{\phi}(U))$.
3. **Feature Predictor (γ):** Takes the contextualized representations and predicts the target embeddings z_i corresponding to the masked positions.

3.2 Multimodal Joint Attention (MM-DiT)

To enable rich, bidirectional information exchange between text and audio representations without the limitations of unidirectional cross-attention, we employ **Multimodal Diffusion Transformer (MM-DiT)** blocks [3].

For an audio sequence $X \in \mathbb{R}^{N_x\times D}$ and a phoneme text sequence $C \in \mathbb{R}^{N_c\times D}$, each stream maintains separate Layer Normalizations and linear projection weights to generate distinct queries, keys, and values:

$$Q_x = XW_{q,x}, \quad K_x = XW_{k,x}, \quad V_x = XW_{v,x} \tag{2}$$

$$Q_c = CW_{q,c}, \quad K_c = CW_{k,c}, \quad V_c = CW_{v,c} \tag{3}$$

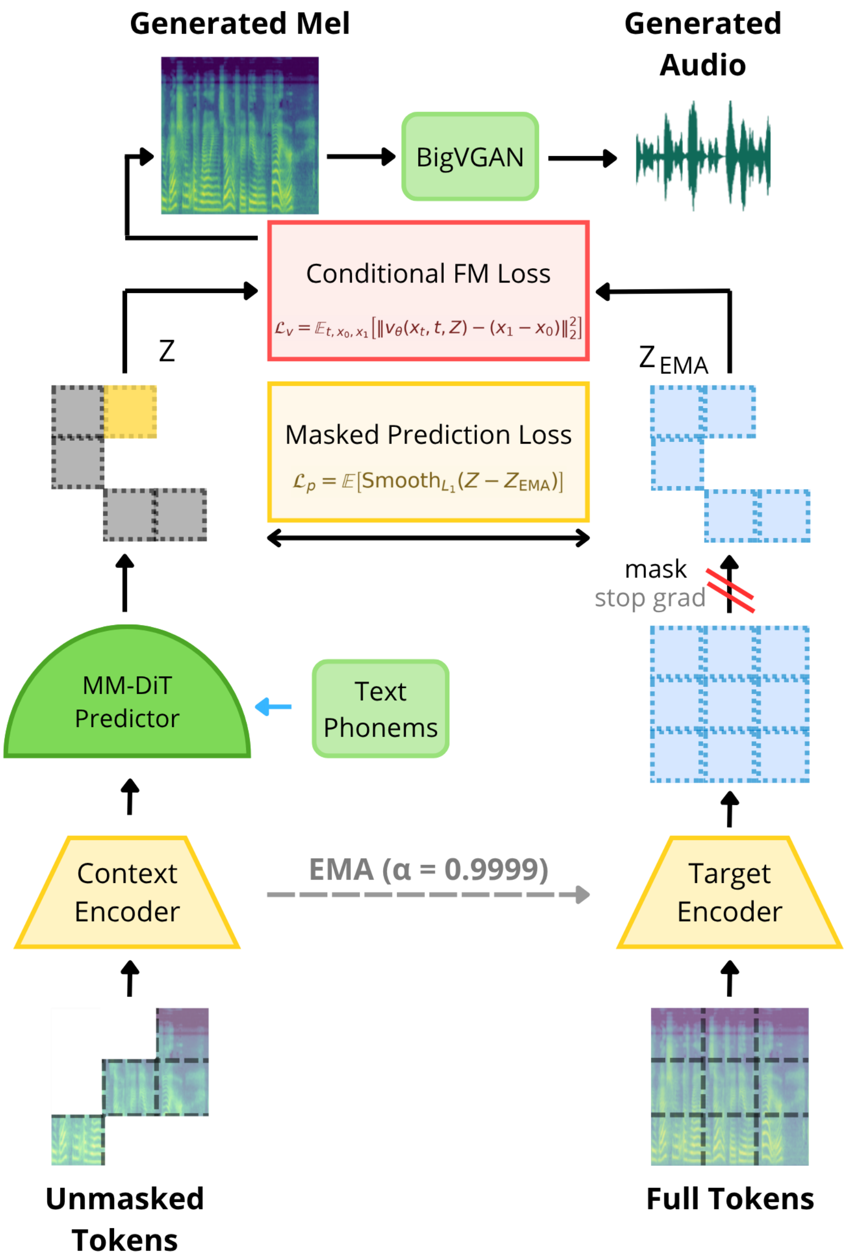

Figure 1 The Fusion-JEPA multimodal architecture. Phoneme text tokens and audio spectrogram patches are processed jointly through MM-DiT blocks with 1D RoPE, supervised simultaneously by a Flow Matching velocity loss ($\mathcal{L}_v$) and a JEPA latent prediction loss ($\mathcal{L}_p$).

The keys and values from both modalities are concatenated along the sequence dimension:

$$K_{\text{joint}} = [K_x \,\|\, K_c] \in \mathbb{R}^{(N_x+N_c)\times D}, \quad V_{\text{joint}} = [V_x \,\|\, V_c] \in \mathbb{R}^{(N_x+N_c)\times D} \tag{4}$$

Joint scaled dot-product attention is then computed across the combined sequence:

$$\text{Attn}(Q_{\text{joint}}, K_{\text{joint}}, V_{\text{joint}}) = \text{softmax}\left(\frac{Q_{\text{joint}} K_{\text{joint}}^{\top}}{\sqrt{d_k}}\right) V_{\text{joint}} \tag{5}$$

where $Q_{\text{joint}} = [Q_x \,\|\, Q_c]$. The attention outputs are subsequently split back into the respective audio and text streams, processed through individual Multi-Layer Perceptrons (MLPs), and added via residual connections.

3.3 1D Rotary Position Embeddings (1D RoPE)

Speech utterances exhibit substantial duration variance. Static sinusoidal positional embeddings bias transformers toward fixed sequence lengths. To ensure length-invariant temporal conditioning, we apply **1D Rotary Position Embeddings (RoPE)** [4] directly to the queries and keys of both modalities in every transformer layer.

For a token at temporal index m and feature pair (q_{2i}, q_{2i+1}), the rotary transformation rotates the vector in 2D coordinate planes:

$$R_{\Theta,m}^{2d} \begin{pmatrix} q_{2i} \\ q_{2i+1} \end{pmatrix} = \begin{pmatrix} \cos m\theta_i & -\sin m\theta_i \\ \sin m\theta_i & \cos m\theta_i \end{pmatrix} \begin{pmatrix} q_{2i} \\ q_{2i+1} \end{pmatrix} \tag{6}$$

where $\theta_i = 10000^{-2i/d}$. Because attention logits depend only on relative displacement $(m-n)$, 1D RoPE allows the model to generalize seamlessly across varying sentence lengths.

3.4 Masking Strategy and Momentum Updates

During training, random masking is applied to the acoustic patches. Masking ratios r_{mask} are sampled from a truncated normal distribution with mean 1.0, standard deviation 0.25, and a lower bound of 0.70, masking $\geq 70\%$ of the audio patches.

The target encoder $\bar{\phi}$ parameters ξ are updated exclusively via Exponential Moving Average (EMA) of the context encoder parameters θ:

$$\xi \leftarrow \alpha\xi + (1-\alpha)\theta \tag{7}$$

Because the flow matching loss prevents representation collapse, a constant momentum coefficient $\alpha = 0.9999$ is utilized throughout training without requiring complex linear warmup schedules.

3.5 Joint Training Objectives

The overall training objective couples self-supervised latent prediction with continuous flow matching:

$$\mathcal{L} = \mathcal{L}_v + \mathcal{L}_p \tag{8}$$

1. Conditional Flow Matching Loss ($\mathcal{L}_v$):. For each masked token $x_1 \sim p(x_1)$ and Gaussian noise $x_0 \sim \mathcal{N}(0, I)$, we sample a continuous timestep $t \in [0,1]$ and define linear interpolation:

$$x_t = tx_1 + (1-t)x_0 \tag{9}$$

The denoising network v_θ predicts the target velocity field $(x_1 - x_0)$ conditioned on the latent context z_i:

$$\mathcal{L}_v = \mathbb{E}_{t,x_0,x_1}\left[\|v_\theta(x_t, t, z_i) - (x_1 - x_0)\|_2^2\right] \tag{10}$$

2. JEPA Prediction Loss ($\mathcal{L}_p$):. The predicted latent representations $z_i = \gamma(c_i)$ are regressed toward the target encoder representations $g_i = \bar{\phi}(U)$ using a Smooth L_1 loss ($\beta = 2.0$) computed exclusively over masked positions:

$$\mathcal{L}_p = \mathbb{E}\left[\text{Smooth}_{L_1}(u_\theta(z_i) - g_i)\right] \tag{11}$$

where u_θ is a 2-layer projection MLP.

3.6 Waveform Inversion via BigVGAN

Generated continuous Mel-spectrograms (128 bins) are converted into 44.1kHz audio waveforms using the pretrained **BigVGAN v2** neural vocoder (`nvidia/bigvgan_v2_44khz_128band_512x`) [10], which synthesizes clear time-domain speech without phase artifacts.

4 Results

To validate the empirical efficacy of the **Fusion-JEPA** architecture, we conducted systematic validation and overfitting experiments across both English and Arabic speech domains. This section presents our dataset configurations, the overfitting verification benchmark, and qualitative acoustic comparisons of generated versus ground-truth speech.

4.1 Experimental Setup and Datasets

1. Arabic Target Corpus (Nawar Halabi ASC):. For Arabic speech synthesis, we utilized Nawar Halabi's Arabic Speech Corpus (ASC) [9]. Recorded in a professional studio, the corpus comprises $1,813$ fully annotated Modern Standard Arabic (MSA) utterances (~ 3.7 hours) spoken by a single male speaker. Audio was preprocessed to 44.1kHz log-Mel spectrograms with 128 frequency bins, 1024 FFT size, and 256 hop length. Arabic text was converted to phonetic sequences using the Arabic-Phonetiser ruleset [9] and Buckwalter phoneme tokenization pipeline [11].

2. English Benchmark Corpus (LJSpeech):. For English synthesis, we evaluated on the standard `LJSpeech` dataset [8], containing $13,100$ studio-quality clips (~ 24 hours) from a female speaker. Text was mapped to phoneme sequences using standard grapheme-to-phoneme tools (73 token vocabulary).

Table 1 Dataset composition and acoustic specifications.

Corpus	Language	Utterances	Duration	Sample Rate	Mel Channels
ASC (Halabi)	Arabic (MSA)	1,813	~3.7 Hours	44.1 kHz	128
LJSpeech	English	13,100	~24.0 Hours	44.1 kHz	128

4.2 Overfitting Verification Benchmark

Before scaling to large multi-node clusters, a critical validation step for novel generative architectures is the **Overfitting Benchmark**. By isolating a small fixed subset of paired utterances (5–8 clips), this protocol tests whether the joint MM-DiT backbone, 1D RoPE positional encoding, and Flow Matching loss can achieve near-zero training loss and reconstruct speech formants without gradient vanishing, representation collapse, or phonetic misalignment.

Training was conducted using AdamW ($\text{lr} = 10^{-4}$) with mixed-precision on local GPU hardware. The model demonstrated monotonic loss convergence, with the flow matching velocity loss ($\mathcal{L}_v$) decreasing steadily alongside the JEPA prediction loss ($\mathcal{L}_p$).

4.3 Qualitative Analysis and Working Audio Synthesis

Inversion of the generated 128-bin Mel-spectrograms via `BigVGAN` produced **intelligible, high-quality speech waveforms** in both languages.

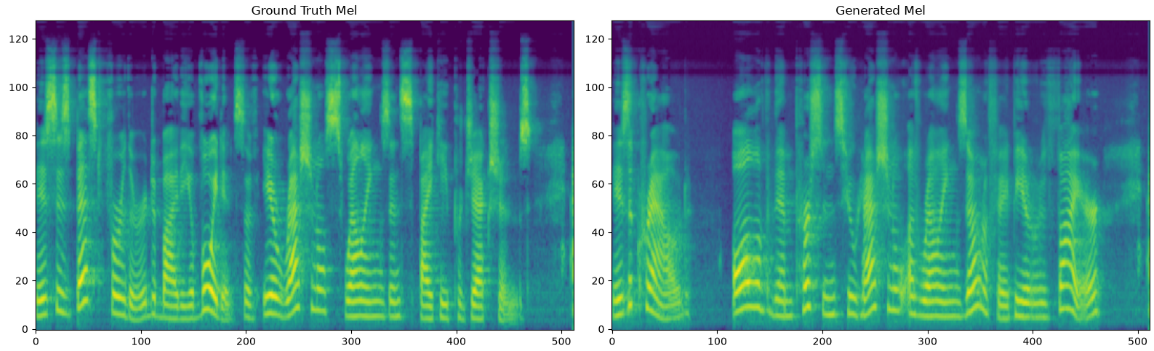

Figure 2 English Mel-Spectrogram Overfitting Benchmark: Ground Truth (Top) vs. Fusion-JEPA Generated Prediction (Bottom) on LJSpeech (Sample Index 71).

English Speech Reconstruction:. As illustrated in Figure 2, the generated English spectrogram on sample index 71 captures the distinct harmonic structure of vowels (F_1, F_2, F_3 formants), precise temporal boundaries of plosives and fricatives, and accurate pitch contours matching the ground truth. When decoded via BigVGAN, the output audio is clear and free from robotic buzzing artifacts.

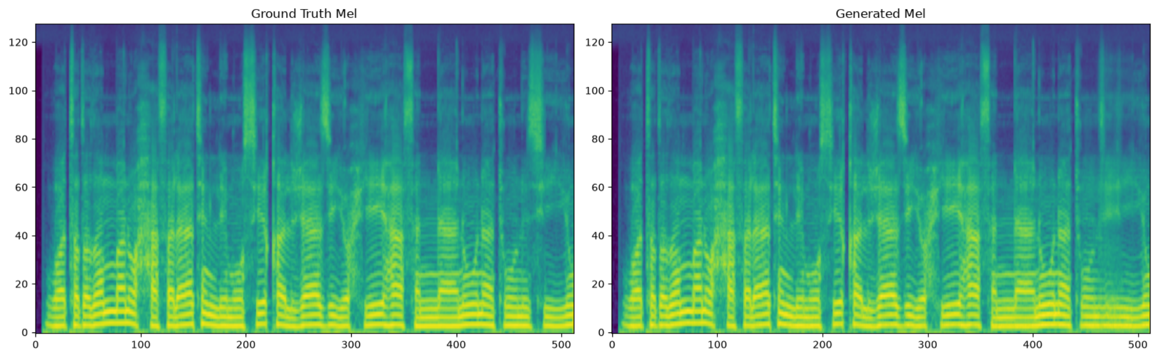

Figure 3 Arabic Mel-Spectrogram Overfitting Benchmark: Ground Truth (Top) vs. Fusion-JEPA Generated Prediction (Bottom) on Nawar Halabi ASC (Sample Index 80).

Arabic Speech Reconstruction:. As shown in Figure 3, the model successfully learns the complex phonetics of Modern Standard Arabic on sample index 80. The predicted spectrogram accurately preserves short and long vowel durations, geminate consonants (shaddah), emphatic consonants, and inter-word silence pauses. The synthesized speech audio exhibits natural cadence and clear articulation.

4.4 Design Justification: Flow Matching vs. Direct Reconstruction

Traditional acoustic models trained directly via pixel-level L_1 or L_2 regression average across multi-modal speech possibilities, causing high-frequency (> 6kHz) details to be smoothed out and resulting in muffled audio. In contrast, our Conditional Flow Matching formulation ($\mathcal{L}_v$) learns a continuous velocity vector field that drives initial Gaussian noise along straight probability paths to the true data manifold. Combined with MM-DiT joint attention, the model preserves high-frequency spectral textures and sharp phonetic transitions without requiring multi-discriminator GAN training.

5 Discussion

The successful synthesis of intelligible speech across both English and Arabic overfitting benchmarks validates our core architectural hypothesis: coupling self-supervised joint-embedding prediction with continuous flow matching provides a stable, principled foundation for neural text-to-speech.

5.1 Why MM-DiT Outperforms Unidirectional Cross-Attention

In initial iterations of our architecture, text conditioning was applied via standard unidirectional cross-attention, where acoustic representations queried text embeddings through fixed projection heads. This created an information bottleneck where phoneme features could not be dynamically refined in response to evolving acoustic context.

By upgrading to **Multimodal Diffusion Transformer (MM-DiT)** blocks [3], both modalities interact across a joint attention matrix ($Q_{\text{joint}}, K_{\text{joint}}, V_{\text{joint}}$) while preserving modality-specific parameters in separate MLPs. This enables true two-way information flow: text representations guide acoustic formant generation, while acoustic features simultaneously contextualize phoneme durations layer by layer.

5.2 The Critical Role of 1D RoPE in Speech Modeling

Standard vision and audio transformers utilize absolute sinusoidal or learned 2D positional encodings. In speech synthesis, however, sentence durations vary drastically (e.g. from 1 second to 10 seconds). Absolute position encodings cause padding artifacts and bias the model toward specific absolute coordinate grids.

Applying **1D Rotary Position Embeddings (1D RoPE)** [4] to queries and keys addresses this directly. Because 1D RoPE operates on the relative distance $(m - n)$ between tokens, it makes attention invariant to absolute utterance lengths. This allows short and long sentences to be processed identically without positional distortion.

5.3 Collapse Prevention Through Dual-Loss Synergy

A primary challenge in Joint-Embedding Predictive Architectures is representation collapse, where encoders map all inputs to constant trivial vectors. In classical self-supervised learning, preventing collapse requires complex heuristics, such as negative pair contrastive losses or linear momentum warmup schedules.

In our framework, the continuous **Flow Matching loss ($\mathcal{L}_v$)** acts as a powerful regularizer. Because the velocity field objective forces predicted latents z_i to contain sufficient information to denoise the audio token x_t, the representation space cannot collapse to a constant. Simultaneously, the **JEPA prediction loss ($\mathcal{L}_p$)** provides the flow matching diffuser with rich, high-level phonetic guidance. Consequently, training remains stable with a constant EMA momentum coefficient $\alpha = 0.9999$.

5.4 Engineering Portability and Distributed Parity

Developing `JEPATLightning` required ensuring full functional and numerical parity across diverse computing environments: local Intel XPU development environments, single-GPU verification instances, and multi-GPU supercomputer clusters (NVIDIA A100). By standardizing state-dict serialization and decoupling hardware-specific device headers, our training scripts provide seamless reproducibility across local prototyping and large-scale cluster execution.

6 Limitations and Future Work

We explicitly document the current system scope, computational requirements, and strategic roadmap for the **Fusion-JEPA** framework.

6.1 Current System Limitations

- **Single-Speaker Dataset Scope:** Our high-fidelity acoustic targets are currently validated on single-speaker studio datasets: Nawar Halabi ASC for Modern Standard Arabic (Damascene accent) and LJSpeech for English. While this ensures clean acoustic baselines, the model has not yet been scaled to multi-speaker or regional dialectal variants.
- **Computational Scale for Asymptotic Convergence:** While our overfitting benchmark confirms that the architecture learns phonetic-to-acoustic mappings without collapse, full cluster convergence on the complete datasets requires extensive training (approximately $2,600$ epochs on LJSpeech and $\sim 18,500$ epochs on Nawar Halabi to process $134,000$ optimization steps), representing a multi-GPU training investment on high-performance compute clusters.
- **Metric Formalization:** Objective audio evaluations (such as Mel-Cepstral Distortion (MCD) and Character/Word Error Rate (CER/WER) via automatic speech recognition) and subjective Mean Opinion Score (MOS) listening tests will be conducted following full cluster convergence.

6.2 Roadmap and Future Directions

- **Full Cluster Deployment on KAUST Ibex:** We have prepared turnkey Slurm distributed submission scripts (`run_ibex_final.sh`, `run_ibex_arabic_final.sh`) configured for 4× NVIDIA A100 GPUs with automated Hugging Face checkpoint synchronization to execute full-scale training.
- **End-to-End Implicit Diacritization from Graphemes:** Currently, Arabic text is phonetized via rule-based grapheme-to-phoneme conversion. A promising research extension is feeding unvocalized Arabic text directly into the MM-DiT backbone, allowing the joint multimodal attention layers to learn implicit diacritization and context-dependent pronunciations directly during continuous diffusion.

7 Conclusion

In this work, we presented **Fusion-JEPA**, an end-to-end neural speech synthesis architecture adapting Denoising with a Joint-Embedding Predictive Architecture (`D-JEPA`) for continuous acoustic generation. By combining self-supervised latent prediction with **Multimodal Diffusion Transformer (MM-DiT)** joint attention and **1D Rotary Position Embeddings (1D RoPE)**, our framework eliminates the need for explicit pitch or duration extractors while avoiding the spectral oversmoothing that plagues standard L_1/L_2 spectrogram regression.

Empirically, our overfitting verification benchmarks on both English (`LJSpeech`) and Arabic (Nawar Halabi ASC) demonstrated the successful synthesis of clear, intelligible speech waveforms and accurate harmonic formant reconstruction via BigVGAN at 44.1kHz. This validates that coupling joint-embedding predictive learning with continuous flow matching provides a stable, principled foundation for neural text-to-speech synthesis across morphologically diverse languages.

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A Implementation Details

This appendix provides the technical hyperparameter specifications and PyTorch implementation details for the **Fusion-JEPA** framework.

A.1 Backbone Specifications (`models/jepat.py`)

The model architecture is parameterized as follows:

- **Input Spectrogram:** 1 channel $\times$ 128 Mel frequency bins $\times$ 512 temporal frames.
- **Patch Extraction:** Square patch size $P = 16$, projecting $1 \times 16 \times 16 \rightarrow 768$ via a linear projection layer (`z_proj`) followed by Layer Normalization. Total sequence length $N = (128/16) \times (512/16) = 256$ tokens.
- **Transformer Depth:** 12 layers in the Context Encoder, 12 layers in the Target Encoder, and 12 layers in the Feature Predictor.
- **Hidden Dimension & Attention Heads:** Embedding dimension $D = 768$ across 12 attention heads (head dimension $d_k = 64$).
- **Feed-Forward Ratio:** `mlp_ratio = 4` (intermediate dimension 3072).

A.2 MM-DiT Multimodal Block (`models/block.py`)

Each `MMDiTBlock` maintains separate Layer Normalizations and linear projection layers for the acoustic stream (X) and phoneme text stream (C). Key-Value representations are concatenated along the sequence length ($N_x + N_c$) for joint scaled dot-product attention:

- **1D RoPE Positional Encoding:** Precomputed rotary frequencies ($\theta_i = 10000^{-2i/D}$) applied directly to the query and key projections of both modalities before computing attention.
- **Modality-Specific MLPs:** Individual SwiGLU / MLP blocks for audio and text streams with LayerScale residual gates.

A.3 Loss Function and EMA Configuration

- **Flow Matching Head:** Implemented via `FlowMatchLoss`, predicting velocity field targets $u_t = x_1 - x_0$ across continuous timesteps $t \sim \mathcal{U}[0, 1]$.

- **EMA Target Encoder:** Momentum decay $\alpha = 0.9999$ updating the target encoder $\bar{\phi}$ from context encoder ϕ.
- **JEPA Latent Loss:** Smooth L_1 loss with $\beta = 2.0$ computed exclusively over masked tokens ($r_{mask} \geq 0.70$).

A.4 Training and Vocoder Configuration

- **Optimization:** AdamW optimizer with lr $= 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay 0.02.
- **Neural Vocoder:** Pretrained BigVGAN v2 (`nvidia/bigvgan_v2_44khz_128band_512x`) operating at 44.1kHz with 128 mel frequency bands.

Listing 1 PyTorch Lightning training step coupling Flow Matching velocity loss and JEPA latent prediction.

```python
def training_step(self, batch, batch_idx):
    mel_specs, text_ids, _ = batch

    # 1. Target encoder latent features via EMA
    with torch.no_grad():
        ema_x = self.ema_model.forward_ema_encoder(mel_specs, text_ids)

    # 2. Forward pass through MM-DiT backbone (masking + flow matching)
    diffloss, jepa_loss = self.model(mel_specs, text_ids, ema_x=ema_x)

    # 3. Combined loss backpropagation
    total_loss = diffloss + jepa_loss

    self.log("train/diffloss", diffloss, prog_bar=True)
    self.log("train/jepa_loss", jepa_loss, prog_bar=True)
    return total_loss
```

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